

MOTIVATION

Galois $K/\mathbb{Q}$ compute $\text{Gal}(K/\mathbb{Q})$

K any field $G_K = \text{Gal}(K^{\text{sep}}/K)$

Inverse Galois problem

G finite group what is $\mathbb{Q}$ field K
such that $G \cong \text{Gal}(K/\mathbb{Q})$

G profinite group find K

if $|G_K| = 1$ then $K = K^{\text{sep}}$

$|G_K| = 2$ then K is real closed

- -1 is not a sum of squares (real)
- maximal real

examples $\mathbb{R}$, $\mathbb{R} \cap \mathbb{Q}^{\text{sep}}$, $\bigcup_{n \in \mathbb{N}} \mathbb{R}((t^{1/n}))$

in general if $G_K \cong G_L$ for
free is no hope to have $L \cong K$

but we could have $L \cong K$

K and L share the same 1st order properties

not 1st order $\rightarrow$ cardinality

$\rightarrow$ Archimedean property

$\forall x, y \in \mathbb{R} \exists x < y \exists n \in \mathbb{N}$ such

not $y < nx$

(Königsmann) if $G_K \cong G_{\mathbb{Q}_p}$
then $K \cong \mathbb{Q}_p$

example $\mathbb{Q}^{\text{sep}} \cap \mathbb{Q}_p = \bigcup_n \mathbb{Q}_p((t^{1/n}))$

Artin/Schreier / Ax-Kochen

$K \cong \mathbb{Q}_p$ iff K admits a valuation v st.
 $\text{char}(K) = 0$
 $K_v \cong \mathbb{F}_p$
 $v(p)$ is min positive (e.g. $\mathbb{Q}$)

K is maximal such

$\Leftrightarrow v$ is henselian

$\bullet v(K/(v(p)))$ is divisible

(equivalently $v(K) \cong \mathbb{Z}$)

e.g. $\mathbb{Q} \times \mathbb{Z} \cong \mathbb{Z}$

actually all of these are

smth divisible $\times \mathbb{Z}$

Goal of today: prove that $G_{\mathbb{Q}_p}$ is "small"
i.e. $\mathbb{Q}_p$ has finitely many alg. extensions
of deg $n \forall n$

($\rightarrow$ Corollary 3.17)

DEFINITION 3.1

F a field. We denote by

$$\mu_F = \{x \in F \mid x^n = 1 \text{ for some } n\}$$

$$\mu_n = \{n\text{-th root of unity in } F^{\text{alg}}\}$$

ζ_n = primitive n -th root of unity

$$\mathbb{Q}_p^\times = \bigcup_{n \geq 1} \mathbb{Q}_p^{\times n}$$

we always assume compatibility $\mathbb{Q}_p^{\times q} = \mathbb{Q}_p^q$

LEMMA 3.2

$L = \mathbb{Q}_p(\zeta_n)$ • if $(n, p) = 1$ then $L/\mathbb{Q}_p$ is
unramified and

$$\text{Gal}(L/\mathbb{Q}_p) \cong \text{Gal}(\mathbb{F}_p(\zeta_n)/\mathbb{F}_p)$$

$$\cong \mathbb{Z}_k$$

where $k = \text{ord}_n(p)$

(as $p \in (\mathbb{Z}/n\mathbb{Z})^\times$)

(cyclic group
of order k)

• if $n = p^m$ then $L/\mathbb{Q}_p$ is totally ramified
of degree $(p-1)p^{m-1}$

$$\text{and } \text{Gal}(L/\mathbb{Q}_p) \cong (\mathbb{Z}/p^n\mathbb{Z})^\times$$

COROLLARY

if $n \mid p-1 \Rightarrow \zeta_n \in \mathbb{Q}_p$ for p odd

$$\mu_{\mathbb{Q}_p} = \begin{cases} \mu_{p-1} & \text{if } p \text{ odd} \\ \mu_2 & \text{if } p = 2 \end{cases}$$

LEMMA 3.4

let (K, v) henselian valued field
and let q be a prime

Assume (i) $\text{char}(K) \neq q$

(ii) $v|_K$ is not q -divisible

$$(iii) \text{Gal}(K_v(q)/K_v) \cong \mathbb{Z}_q$$

$K_v(q)$ is h.c. q -closure

$\Rightarrow \mathbb{Q}_p$ is fine
and also
all finite
extensions of $\mathbb{Q}_p$

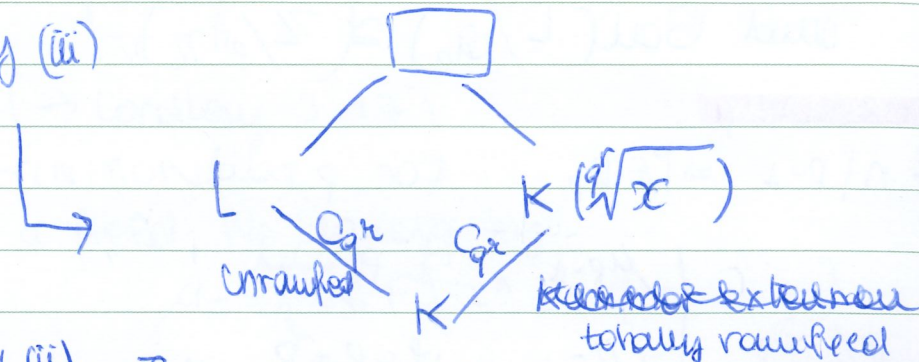
Then $\zeta_{q^k} \in K$ iff K admits a
 $C_{q^k} \times C_{q^k}$ extension

proof

$\Rightarrow$ Assume $\rho_{qr} \in K$

by (ii)

this is our extension



by (ii) with
for $x \in K$, $\rho(x) \in K$ not q -divisible

by Kummer theory

$\Leftarrow$ Let L/K be a $C_{qr} \times C_{qr}$ extension

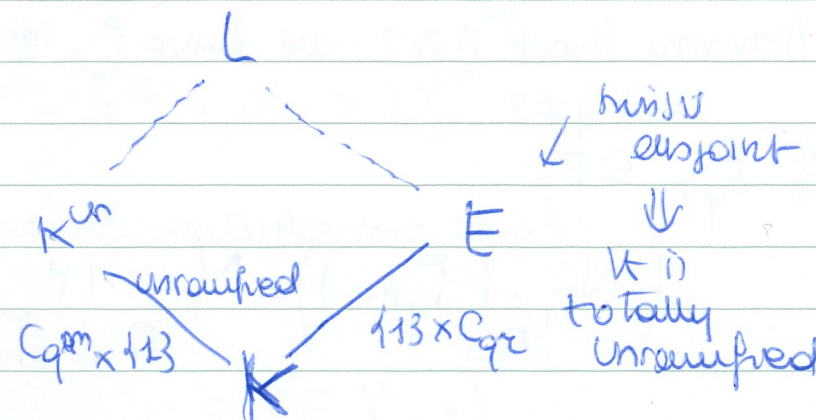
Let $K^{ur} =$ maximal unramified ex in L

$\Rightarrow$ it is a q -extension

all extensions of degree a power of q
are cyclic by (iii)

because the Galois group is a quotient
of $C_{qr} \times C_{qr}$

$\Rightarrow$ we are inside one component of $Q_r \times Q_r$



$$\Rightarrow \text{Gal}(E/K) \cong I(E/K) \cong C_{qr} \\ \cong \text{Hom}(\bigvee_{v \in K} \mu_{E_v})$$

we have
 q -torsion $\Rightarrow \rho_{qr} \in \mu_{E_v}$
(since $E_v = K_v$)

LEMMA 3.6

Let F be a field with $\text{char } F \neq q$

Then $F(\rho_{q^\infty})/F$ is either infinite

in which case $\text{Gal}(F(\rho_{q^\infty})/F) \cong \mathbb{Z}_q \times C$

where C is cyclic of order dividing $q-1$

or it is finite in which case $F(\rho_{q^\infty}) = F(\rho_q)$
for q odd and $F(\rho_{2^\infty}) = F(\rho_4)$ otherwise

proof $F_n = F(\zeta_{q^n})$ for $n \in \mathbb{N}$

Assume that ^{for} $n \geq 2$ we have $\zeta_{q^n} \notin F_{n-1}$

and $q \neq 2$

if $\zeta_{q^{n+1}} \in F_n$

$$(N_{F_n/F_{n-1}}(\zeta_{q^{n+1}}))^q = N_{F_n/F_{n-1}}(\zeta_{q^n}) = \zeta_{q^{n-1}}$$

with $\zeta_{q^n} \notin F_{n-1}$

$\Rightarrow F_n$ is an infinite degree tower

NEW GOAL

$(F^x)/(F^x)^q$ we want to understand its dimension for $F/\mathbb{Q}_p$ finite

$$F^x \supseteq U_F^{(0)} \supseteq U_F^{(1)} \supseteq \dots \supseteq U_F^{(k)} \supseteq \dots$$

DEF 3.7

$$U_F^{(0)} = \mathcal{O}_F^x$$

$$U_F^{(k)} = \{x \in \mathcal{O}_F^x \mid x \equiv 1 \pmod{\pi^k}\}$$

for $k \geq 1$

~~exp~~

FACT 3.8

$$F^x = \pi^{\mathbb{Z}} \times \mu_F^1 \times U_F^{(1)}$$

$\mu_F^1 = \{\text{roots of unity prime to } p\}$

FACT 3.9

$$U_F^{(0)}/U_F^{(1)} \cong (F_v)^x, \cdot$$

$$U_F^{(k)}/U_F^{(k+1)} \cong (F_v, +)$$

proof

$$\begin{aligned} x U_F^{(1)} &\mapsto x + M_F \\ (1 + \pi^k a) U_F^{(k+1)} &\mapsto a + M_F \end{aligned}$$

A big proof of a small result (suite & fin)

Cor 3.11 $F/\mathbb{Q}_p$ finite, $\pi \in F$ a uniformizer. Let $A \subseteq \mathbb{Q}_p^\times$

be a lift of $F^\times$. Then any $x \in U_F^{(1)}/U_F^{(p+1)}$

has a unique representation of the form $a U_F^{(p+1)}$

with $a = 1 + a_1 \pi + \dots + a_k \pi^k$ & $a_i \in A$.

[Note: if $F = \mathbb{Q}_p$, we're writing x as $\sum a_i p^i$;
now we're trying to write $x \in F/\mathbb{Q}_p$ as $\sum a_i \pi^i$, up
to some k .

Proof of 3.11 existence is by 3.9, uniqueness is easy.

Now we need some calculations:

Lem 3.12 $F/\mathbb{Q}_p$ finite and assume $\zeta_p \in F$ Then

$$\frac{p}{(\zeta_p - 1)^{p-1}} \equiv -1 \pmod{(\zeta_p - 1)}$$

$$\frac{p}{(\zeta_p - 1)^{p-1}} \text{ has residue } -1 \in F^\times$$

Proof: take $f(x) = x^{p-1} + \dots + x + 1 = \frac{x^p - 1}{x - 1} = \prod_{i=1}^{p-1} (x - \zeta_p^i)$

We have $f(1) = p$, so $\frac{p}{(\zeta_p - 1)^{p-1}} = \frac{f(1)}{(\zeta_p - 1)^{p-1}} = \prod_{i=1}^{p-1} \left(\frac{1 - \zeta_p^i}{\zeta_p - 1} \right) = (-1)^{p-1} \prod_{i=1}^{p-1} \left(\frac{\zeta_p^i - 1}{\zeta_p - 1} \right)$

$$\text{But } \frac{Y_p^i - 1}{Y_p - 1} = 1 + Y_p + \dots + Y_p^{i-1} = 1 + (1 + (Y_p - 1)) + \dots + (1 + (Y_p - 1))^{i-1} \\ \equiv i \pmod{(Y_p - 1)}.$$

$$\text{So } \frac{p}{(Y_p - 1)^{p-1}} \equiv (-1)^{p-1} (p-1)! \pmod{(Y_p - 1)}$$

We have $v(Y_p - 1) = \frac{1}{p-1}$, so $p \in (Y_p - 1)$; and since

$$(-1)^{p-1} \equiv 1 \pmod{p}, \quad (p-1)! \equiv -1 \pmod{p},$$

$$\text{We conclude } \frac{p}{(Y_p - 1)^{p-1}} \equiv -1 \pmod{(Y_p - 1)}.$$

Lemma 3.13 $F/\mathbb{Q}_p(Y_p)$ finite, $e = e(F/\mathbb{Q}_p(Y_p))$. Then

$$U_F^{(ep+1)} \subseteq (U_F^{(e)})^p$$

$$\text{and if } F = \mathbb{Q}_p(Y_p), \text{ then } U_F^{(p+1)} = (U_F^{(1)})^p.$$

Proof: $\mathbb{Q}_p(Y_p)/\mathbb{Q}_p$ is tot. ram. of deg $p-1$, & $\pi: Y_p - 1$ is a uniformizer. Let Π be a unif. for F .

$$\text{We have } v(p) = 1, \quad v(\pi) = \frac{1}{p-1}, \quad v(\Pi) = \frac{1}{e(p-1)},$$

$$\text{so } U_F^{(e)} = 1 + (\pi) \quad \& \quad U_F^{(ep+1)} = 1 + (\pi^p \Pi).$$

For $x = 1 + \pi^p a \in U_F^{(ep+1)}$ with $a \in \Pi$. Let

$$f(x) = \pi^{-p} [(1 + \pi x)^p - (1 + \pi^p a)] = x^p - a + \sum_{i=1}^{p-1} \binom{p}{i} \pi^{i-p} x^i$$

we have $v(a) > 0$, $v\left(\binom{p}{i} \pi^{i \cdot p}\right) = v\left(\binom{p}{i}\right) + (i-p)v(\pi)$

$$= 1 + \frac{i-p}{p-1}$$

$$= \frac{p-1+i-p}{p-1} = \frac{i-1}{p-1} > 0 \quad \text{if } i \geq 2$$

& $v(p \pi^{1-p}) = 0$, $\text{res}(p \pi^{1-p}) = -1$ (by 3.12)

thus $f(x) = X^p - X$, has simple root in F_v ,

& by Hensel f has a root in $\mathcal{O}_F$:

$$\exists b \in \mathcal{O}_F \text{ s.t. } \pi^{-p} \left((1+\pi b)^p - (1+\pi^p a) \right) = 0$$

$$\text{so } x = (1+\pi b)^p \stackrel{=x}{\in} (U_F^{(e)})^p \subseteq U_F^{(e)}$$

If $F = \mathbb{Q}_p(\zeta_p)$, then $\pi = \zeta_p - 1$, & $\forall b \in \mathcal{O}_F$.

$$(1+\pi b)^p \equiv 1 + p\pi b + \pi^p b^p \pmod{\pi^{p+1}}$$

$$\equiv 1 + \pi^p (b^p - b) \pmod{\pi^{p+1}} \quad (\text{since } p\pi = \frac{p}{\pi^{p-1}} \pi^p)$$

But $F_v = \mathbb{F}_p$ & $b^p \equiv b \pmod{\pi}$; thus $(1+\pi b)^p \equiv 1 \pmod{\pi^{p+1}}$

that is, $(U_F^{(e)})^p \subseteq U_F^{(p+1)}$.

The next results is not needed for our goal today, but is useful later.

Lem 3.14 $F/\mathbb{Q}_p(\mathcal{S}_p)$ finite, $\pi \in F$ unif., $e = e(F/\mathbb{Q}_p(\mathcal{S}_p))$, $f = f(F/\mathbb{Q}_p(\mathcal{S}_p))$

Let $\{b_1, \dots, b_f\} \subseteq \mathcal{O}_F$ be a lift of an $\mathbb{F}_p$ -basis of F_v

Then $\{\pi\} \cup \{1 + b_i \pi^k\}_{\substack{1 \leq k \leq e \\ 1 \leq i \leq f}}$ generates $F^\times$ modulo $F^{\times p}$.

If $F = \mathbb{Q}_p(\mathcal{S}_p)$, then it is even a basis.

Proof sketch we have $F^\times/F^{\times p} \cong \pi^\mathbb{Z}/\pi^p\mathbb{Z} \times U_F^{(1)}/(U_F^{(1)})^p$

π generates $\pi^\mathbb{Z}/\pi^p\mathbb{Z}$. We then prove that

$\{1 + b_i \pi^k\}_{\substack{1 \leq k \leq e \\ 1 \leq i \leq f}}$ generates $U_F^{(e)}/U_F^{(e+p)}$

for any e ; proof by induction on e (from $e=1$,

decreasing). For the case $F = \mathbb{Q}_p(\mathcal{S}_p)$, it's

a basis by a counting argument.

We can finally study $F^\times/F^{\times q}$ for any q .

Lem 3.15 (Degree Lemma) $F/\mathbb{Q}_p$ finite of deg n , $\mathcal{S}_q \in F$, q prime.

Then $F^\times/F^{\times q} = \begin{cases} 2 & q \neq p \\ n+2 & q = p \end{cases}$

Proof Recall that $F^\times \cong \pi^\mathbb{Z} \times \mu_F^\times \times U_F^{(1)}$.

If $q \neq p$, then any $x \in U_F^{(1)}$ has a q^{th} -root by Hensel,
 so $U_F^{(1)} / (U_F^{(1)})^q \cong \{1\}$. (because $x \in U_F^{(1)}$
means $\text{res}(x) = 1$)

Lemma 3.2 $\Rightarrow \mu_F^\times = \mu_{p^b-1}$ (actually, this was only
 with $f = f(F/\mathbb{Q}_p)$. done for $F = \mathbb{Q}_p$, but
 can be obtained from 3.4)

$\exists q \in \mu_F^\times$ by assumption, so $q/p \neq 1$ & $\mu_F^\times / \mu_F^{q^2} \cong C_q$.

Since $\pi^\mathbb{Z} / \pi^q \mathbb{Z} \cong C_q$, we get $F^\times / F^{\times q} \cong C_q \times C_q$.

If $q = p$, we have again $\pi^\mathbb{Z} / \pi^p \mathbb{Z} \cong C_p$, & we now

have $\mu_F^\times / \mu_F^{p^2} \cong \{1\}$. Remains to study $U_F^{(1)} / (U_F^{(1)})^{p^2}$.

we have $U_F^{(1)} / (U_F^{(1)})^p \cong \left(U_F^{(1)} / U_F^{(ep+1)} \right) / \left((U_F^{(1)})^p / U_F^{(ep+1)} \right)$ since by 3.13
 $(U_F^{(1)})^p \in U_F^{(ep+1)}$

with $e = e(F/\mathbb{Q}_p(\mathbb{Z}_p))$. So $n = e(p-1)f$. By 3.11:

$$\left| U_F^{(1)} / U_F^{(ep+1)} \right| = |F_v|^{ep} = (p^f)^{ep} = p^{h + \frac{n}{p-1}}.$$

Claim The Frobenius $x \mapsto x^p$ induces a well defined morphism

$$\Phi: U_F^{(1)} / U_F^{(ep+1)} \longrightarrow (U_F^{(1)})^p / U_F^{(ep+1)}$$

& has kernel of size p .

proof of claim Take $\pi \in F$ a uniformizer, then $v(\pi^{e(p-1)}) = 1$.
(& $v(p) = 1$)

Thus $(1 + b\pi^{e+1})^p \equiv 1 \pmod{\pi^{ep+1}} \quad \forall b \in GF,$

That is, $(U_F^{(e+1)})^p \subseteq U_F^{(ep+1)}$, so Φ is well defined & surjective. For the kernel, use 3.11 to represent elmts of $U_F^{(1)} / U_F^{(e+1)}$ as $1 + a_1\pi + \dots + a_e\pi^e$, where $\{a_i\}_i$ is a lift of F_v , & solve $(1 + a_1\pi + \dots + a_e\pi^e)^p - 1 = 0 \pmod{\pi^{ep+1}}$.

(details skipped).

& finally:

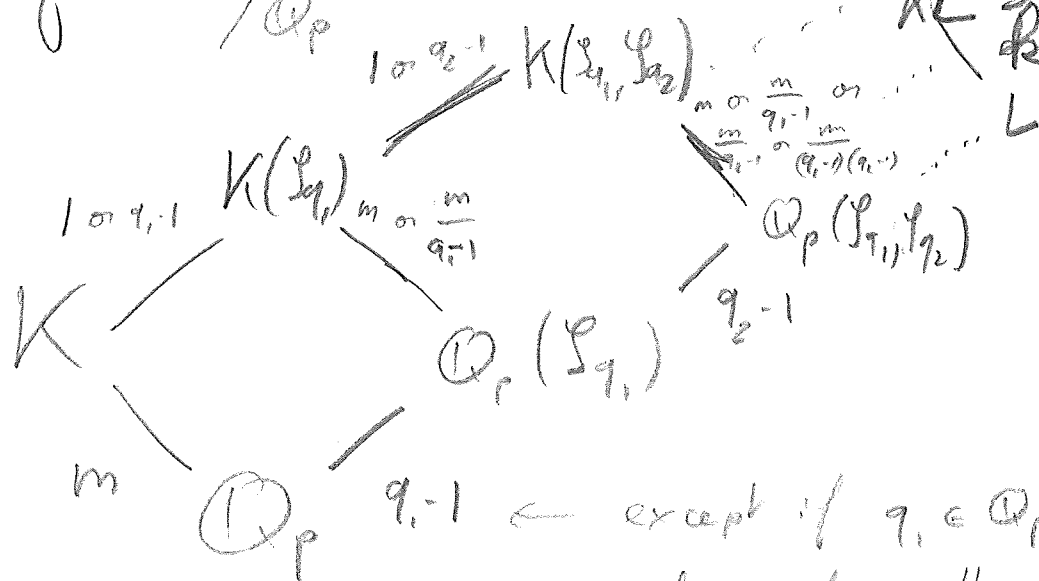
Cor 3.17 $G_{\mathbb{Q}_p}$ is small; $\mathbb{Q}_p$ has finitely many distinct extensions of deg $n \quad \forall n$.

Proof assume $\mathbb{Q}_p$ has infinitely many ext. of deg n . Take the normal hull; some distinct ext could have the same normal hull, but only finitely many such. So, $\mathbb{Q}_p$ has infinitely many distinct _{normal} ext of deg $\leq n$. Pigeonhole $\rightarrow \mathbb{Q}_p$ has infinitely many _{normal} ext of deg m for some m .

Let $L = \mathbb{Q}_p(\zeta_{q_1}, \zeta_{q_2}, \dots, \zeta_{q_r})$ where q_i are all prime divisors of m .

Any extension $F(\mathbb{F}_q)/F$ is of deg 1 or $q-1$, for F of char not q . here $\text{char}(\mathbb{Q}_p) = 0$, of course.

If $K/\mathbb{Q}_p$ is a normal ext of deg m , we have



except if $q_1 \in \mathbb{Q}_p$, but then it doesn't matter.

So the infinitely many normal ext of $\mathbb{Q}_p$ of deg m induce infinitely many normal ext of L of deg dividing m . Some of them might be trivial, but only finitely many; thus, pigeonhole $\rightarrow L$ has infinitely many normal ext of degree k for some $k|m$.

They are solvable (in fact, $G_{\mathbb{Q}_p}$ is pro-solvable), so they are a tower of C_q -extensions; since $L \ni \mathbb{F}_q$, these are Kummer extensions, of the form $L(\sqrt[q]{x})$ for some $x \in L^\times/L^{\times q}$; there's only finitely many of those by Degree Lemma (3.15).